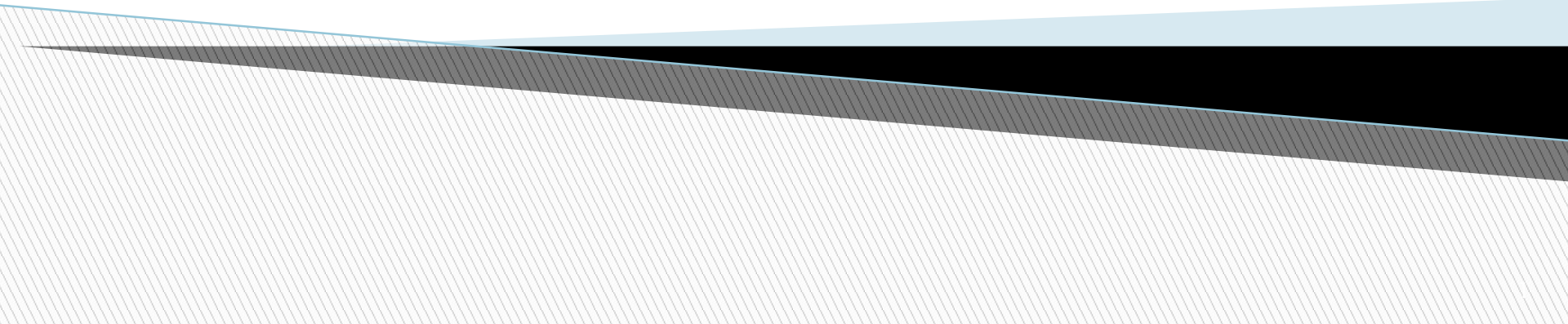


Artificial Bee Colony (ABC) Algorithm



Artificial Bee Colony (ABC) (History)

- Artificial Bee Colony (ABC) is one of the most recently defined algorithms by Dervis Karaboga in 2005, motivated by the intelligent behavior of honey bees.



Artificial Bee Colony

- The ABC algorithm is a **swarm** based **meta-heuristics** algorithm.
- It based on the **foraging** behavior of **honey** bee colonies.
- The artificial bee colony contains **three** groups of bee:
 - Employed bees
 - Onlookers
 - Scouts



Artificial Bee Colony (ABC)

- **The Employed Bee:**

It stays on a food source and provides the neighborhood of the source in its memory.

- **The Onlooker Bee:**

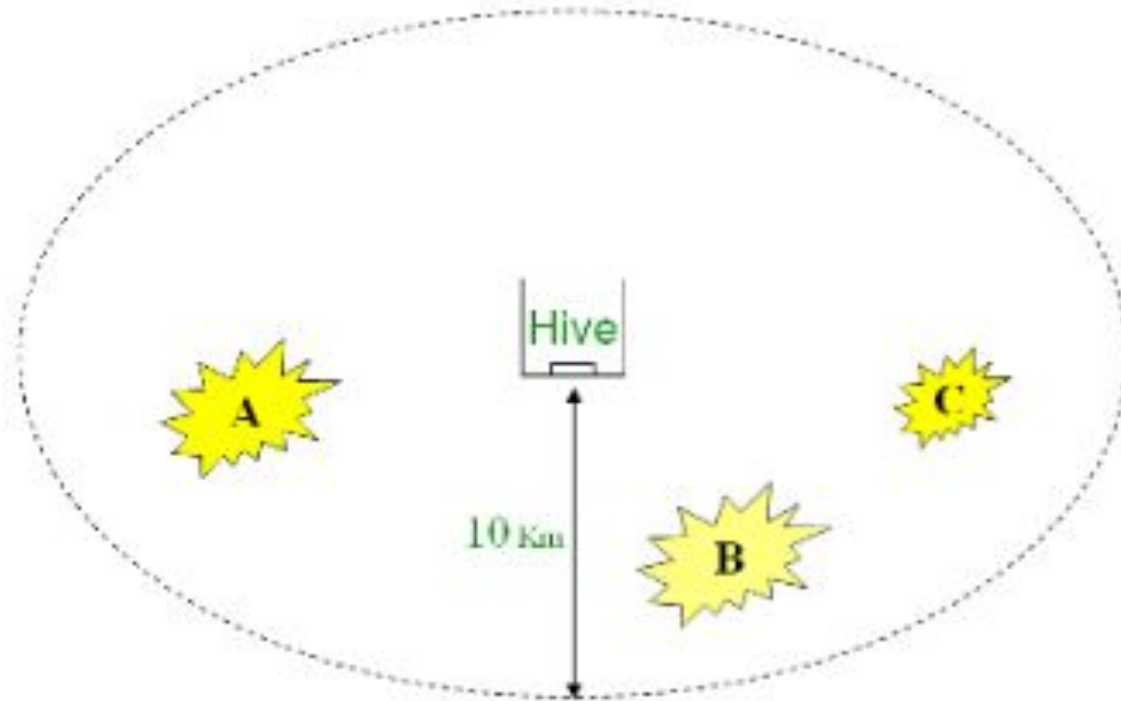
It gets the information of food sources from the employed bees in the hive and select food sources to gathers the nectar.

- **The Scout:**

The employed bee whose food source has been abandoned becomes a scout and starts to search for finding a new food source.

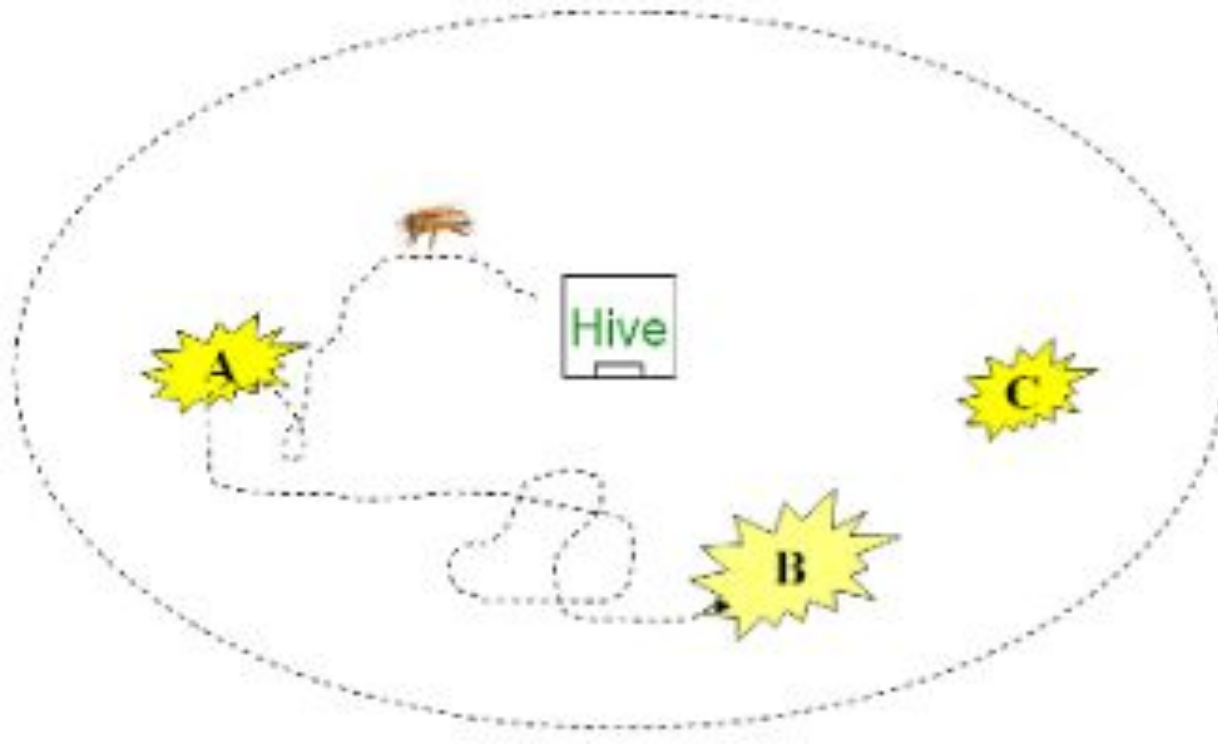
Bees in Nature

- A colony of honey bees can extend itself over long distances in multiple directions (more than 10 km).



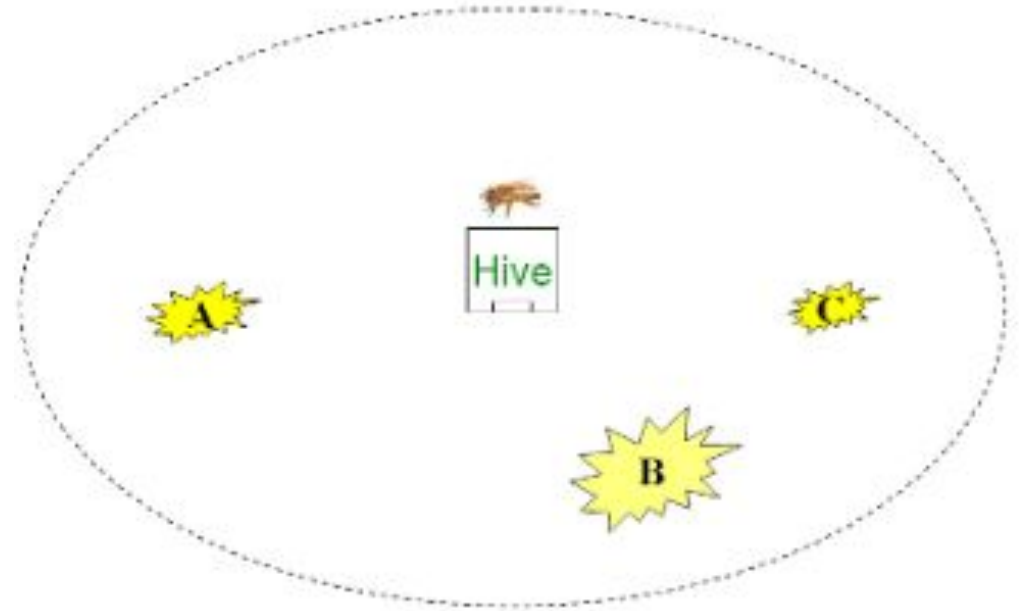
Bees in Nature

- Employed bees go to their food source.



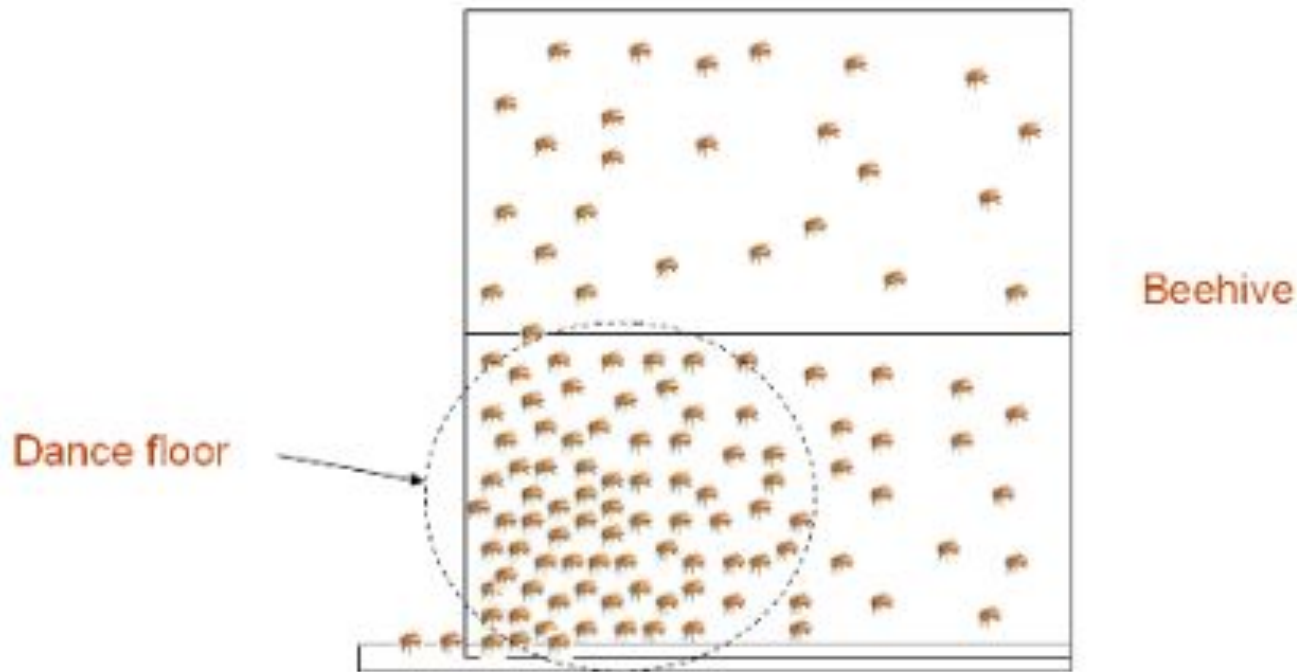
Bees in Nature

- The employed bees who return to the hive, **evaluate** the different patches depending on certain quality threshold (measured as a combination of some elements, such as sugar content).



Bees in Nature

- They go to the “**dance floor**” to perform a “waggle dance”



Bees in Nature

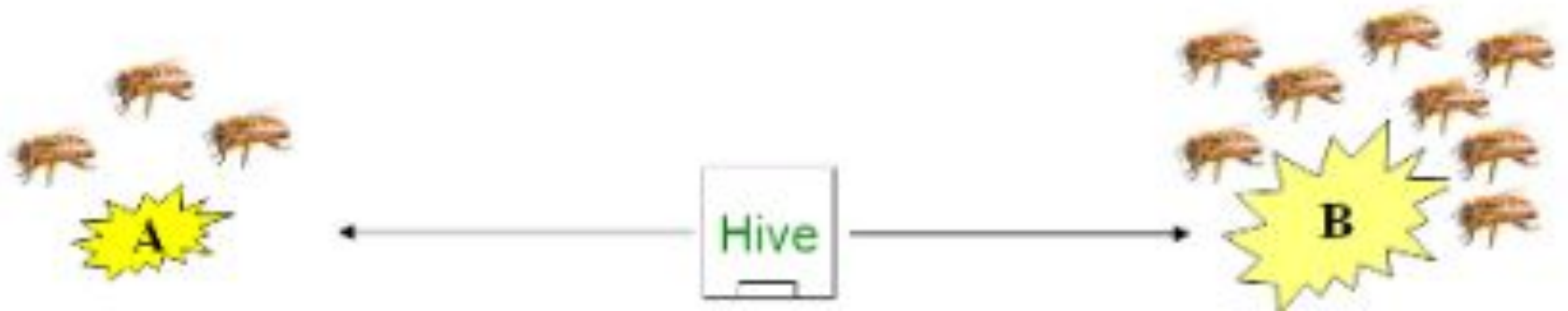
- Bees **communicate** through this **waggle dance** which contains the following information:
 1. The **direction** of flower patches (angle between the sun and the flower patch)
 2. The **distance** from the hive (duration of the dance)
 3. The **quality** rating (fitness) (frequency of the dance)

Bees in Nature

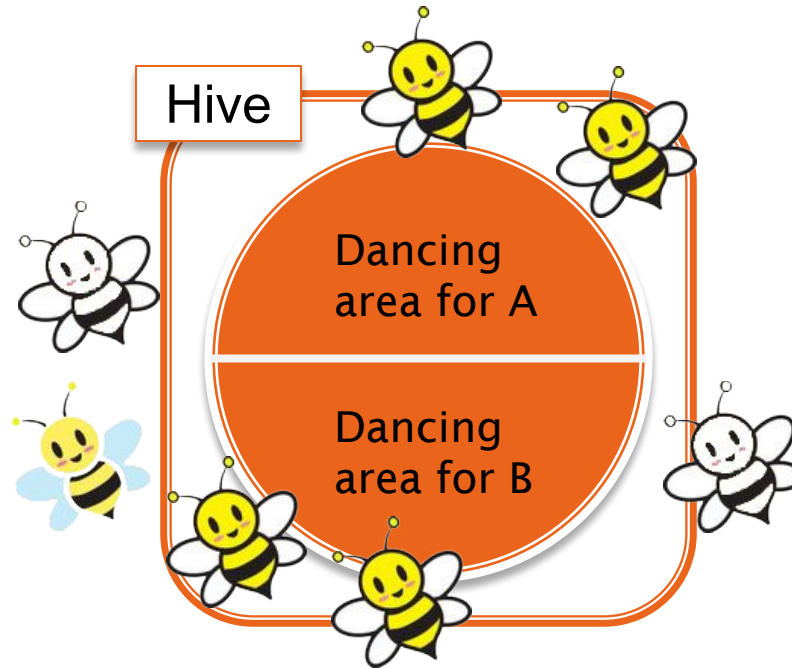
- These information helps the colony to send its bees precisely
- **Follower bees (onlooker bees)** go after the dancer bee (employed bees) to the patch to gather food efficiently and quickly

Bees in Nature

- Thus, according to the fitness, patches can be **visited by more bees** or may be **abandoned**
- The **employed bee** whose food source has been **abandoned** becomes a **scout** and starts to search for finding a **new food source**



Bees in Nature



ABC Algorithm

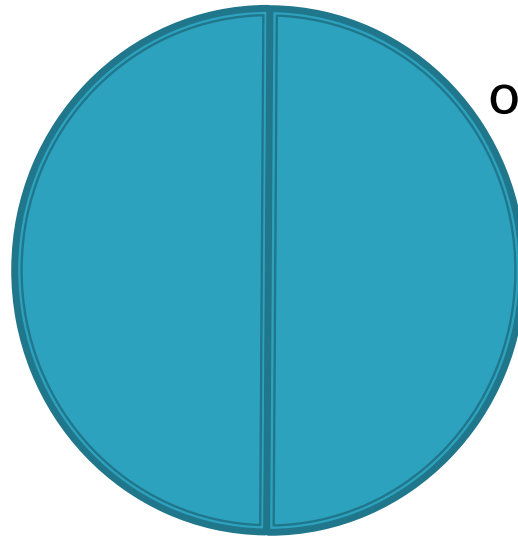
- Step 1: Assign control parameters
- Step 2: Initialize solutions and calculate fitness (employed bees)
- Step 3: Repeat until **stopping criteria** is met
 - employed bees generated neighborhood position and calculate fitness
 - onlooker bees generated position and calculate fitness
 - scout bees generated position and calculate fitness
 - memorize the best solution
- Step 4: Stop

ABC Algorithm

Foraging bee

employed

onlooker



Scout

ABC control parameters

- Swarm size (Total solutions)
- Employed bees (50% of swarm)
- Onlookers (50% of swarm)
- Scouts (1 or can be more than 1)
- Limit (for becoming a scout from an employed bee)
- Dimension of the problem



Initial Solutions

Food source initialization

(Number of solutions = Employed bees)

$$x_{i,j} = x_{\min,j} + rand(0,1)(x_{\max,j} - x_{\min,j})$$

Where,

$i = 1, 2, \dots, n$

n = Food source

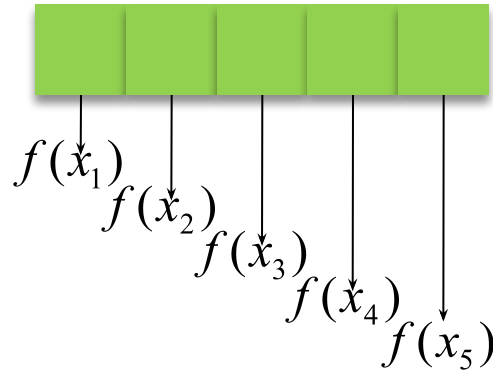
j = Dimension

Employed bee Phase

- Send each employed bee to solutions/positions of flowers (Initial Solutions)
 - Number of solutions = Employed bees = 5 (in this example)
 - 1 dimension (in this example)



- Calculate fitness



Employed bee Phase (Cont.)

- Evolve solution to neighborhood

$$v_{ij} = x_{ij} + \phi_{ij} (x_{ij} - x_{kj})$$

Where,

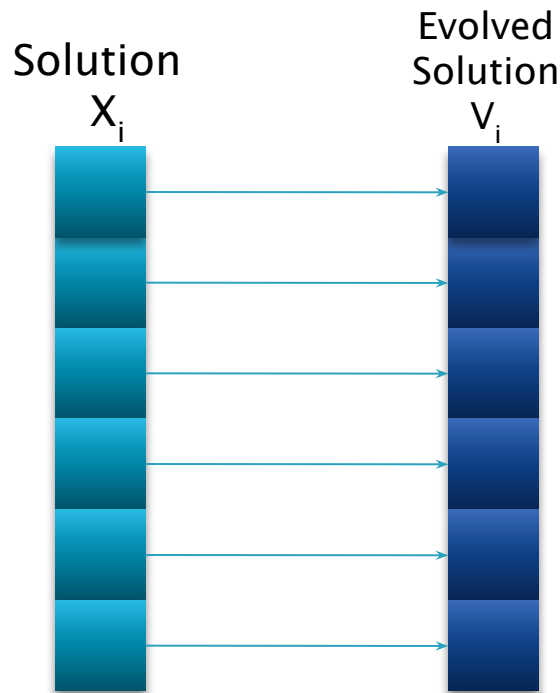
$\phi_{ij} = \text{rand}(-1, 1)$

$i = 1, 2, \dots, n$

n = Food source

j = Dimension

$k = \text{rand}(1, n) \neq i$



$j=6$

- Single solution evolution

- Select better solution comparing $f(x_i)$ and $f(v_i)$

Onlookers Phase

- Calculate probability for each solution

$$P\{x_i\} = \frac{f(x_i)}{\sum_{i=1}^n f(x_i)}$$

- Select solution with respect to probability
- Can use roulette wheel or other technique to make probabilistic nature

Onlookers Phase (cont.)

- Evolve **selected solution**

$$v_{ij} = x_{ij} + \phi_{ij} (x_{ij} - x_{kj})$$

Where,
 $\phi_{ij} = \text{rand}(-1, 1)$
 $i = 1, 2, \dots, n$
 $n = \text{Food source}$
 $j = \text{Dimension}$
 $k = \text{rand}(1, n) \neq i$

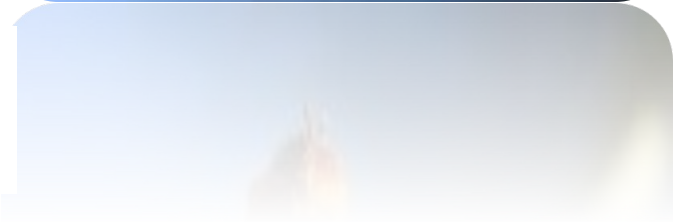
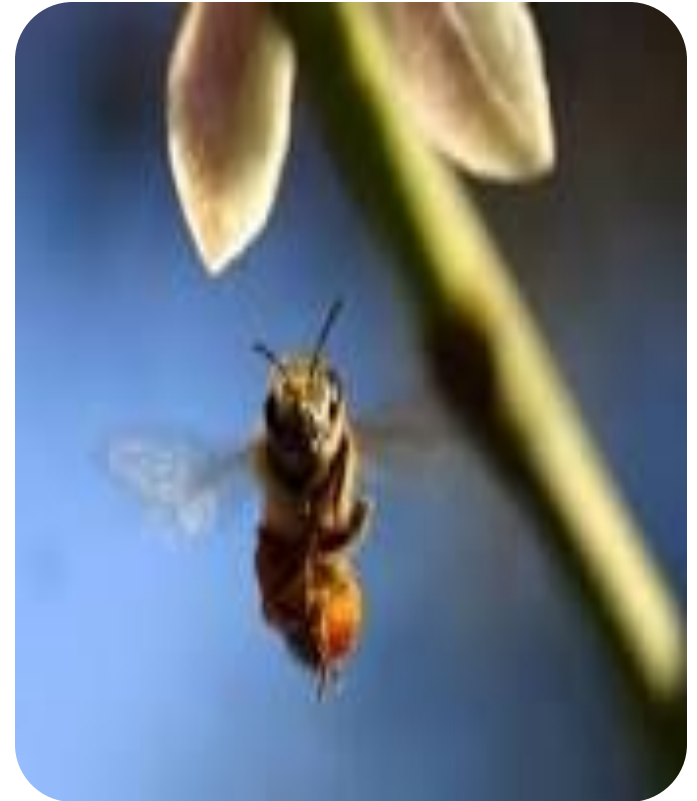
- Calculate $f(v_i)$
- **Select better solution comparing $f(x_i)$ and $f(v_i)$**
(Same as Employed bee phase)

Scout Phase

• In ABC, providing that a position can not be improved further through a predetermined number of cycles, which is called “limit” then that food source is assumed to be abandoned.

-some employed bees (food source abandoned) may become scouts to explore the search space by the following equation (same as initialization)

$$x_i^j = x_{\min}^j + \text{rand}(0, 1)(x_{\max}^j - x_{\min}^j)$$



Advantages & Disadvantages

□ Advantages

- Few control parameters
- Fast convergence
- Both exploration & exploitation

□ Disadvantages

- Search space limited by initial solution

Applications

- Mechanical designs like:
 - Design of welded beam
 - Design of coil spring
- Digital Filter Optimisation
- Fuzzy Control Design
- Data Clustering (solving the local optimum for K-means algorithm)
- Robot control